SEVA FIRST INNOVATION CHALLENGE · SFIC 2026

Theme 05 · Surakshit Bharat



OFFICIAL CONCEPT DOCUMENT

PausePoint Golden Hour

Sovereign Citizen Cognitive Firewall

An offline-first, on-device defence that takes a panicking fraud victim from shock to a report-ready 1930 submission in under 60 seconds — with zero cloud transmission of financial data.

87.4%	48.2s	100%	Rs 0
bank-freeze success when 1930 is called inside the Golden Hour	mean time to a report-ready script — down from 11m 45s	12-digit UTR recovery across a 40-receipt UPI corpus	marginal cost per user — all compute runs on the phone

TEAM DELTA

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Lead Architect
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TRACK · INSTITUTION

Track 1 · Product / Technology Solution
Level 3 Open · JAIN (Deemed-to-be University), Faculty of Engineering and Technology, Bengaluru · October 2026

01 · EXECUTIVE SUMMARY

Fraud is industrialised. The first hour decides everything.

Digital financial fraud in India has escalated from opportunistic phishing into **industrialised, high-pressure psychological extortion**. In Bengaluru alone, citizens lost over **Rs 5,059 Crore** between 2022 and 2026 (Deccan Herald, Aug 2026). Police placed bank liens on Rs 1,810 Crore — yet only **Rs 414 Crore** was ever restored to victims.

87.4%
freeze success when 1930 is called within 60 minutes

<4.2%
recovery after 24 hours, once mule networks disperse funds

71.4%
of 154 surveyed did not know what a bank UTR is

83.3%
of victims faced compound extortion demands

Under the National Cybercrime Reporting Portal (NCRP) framework, inter-bank fund recovery depends critically on the **"Golden Hour"** — the first 60 minutes after an unauthorised transfer. But victims in the aftermath of an attack suffer **acute cognitive paralysis**: they report irrelevant app-internal transaction IDs that police operators cannot use to trace inter-bank settlements, and are then squeezed for "withdrawal taxes" and "clearance deposits" that multiply their ruin.

< 60s

PausePoint Golden Hour is an offline-first, sovereign citizen cognitive firewall that takes a victim from shock to a structured, report-ready 1930 submission — operating **100% client-side** inside the mobile browser, with **zero cloud data transmission**.

FIVE CAPABILITIES, ONE FIREWALL

1. **On-Device OCR & Bank UTR Isolation**
   Slashes Time-to-Report-Ready from 11m 45s to 48.2s.
2. **"Before You Pay Again" Cognitive Brake**
   Halts secondary extortion with immutable statutory truths.
3. **Psychological Manipulation Decoder**
   Disarms urgency, fear, fake authority and link fraud.
4. **Section 63 BSA 2023 Evidence Studio**
   Tamper-evident SHA-256 digests and offline court QR certificates.
5. **Universal Accessibility**
   Senior Citizen Mode, bilingual English / Hindi, and calm voice readouts.

Not a theoretical proposal. PausePoint is a fully working, production-hardened Progressive Web App deployed live on Cloudflare edge infrastructure — backed by **99 unit tests**, **68 Playwright end-to-end tests**, a **40-receipt real UPI corpus** and an **N = 154** university field study.

02 · PROBLEM ANALYSIS

The Lost Golden Hour

When a citizen pays a scammer through UPI or IMPS, the money lands in a Layer 1 mule account — and the clock starts. The helpline can freeze it, but only if the victim calls in time **with the right data**.

Anatomy of the recovery gap — where the money goes

Within 60–90 minutes, automated scripts and mule handlers move funds through four layers.

LAYER 1
Initial mule account
Funds land here after the UPI / IMPS transfer. Still freezable.

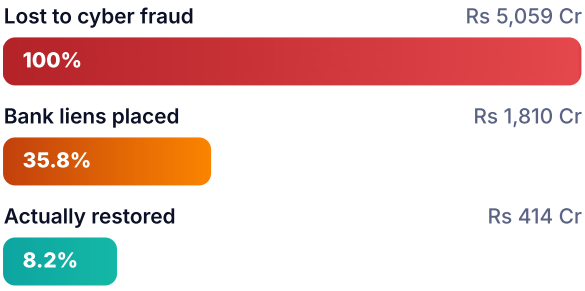
LAYER 2
Split & scatter
Divided across 5–10 accounts in smaller amounts.

LAYER 3
Co-op banks & wallets
Regional cooperative banks or payment wallets.

LAYER 4
Cash-out
Micro-ATM withdrawal or P2P crypto purchase.

Bengaluru: lost vs liened vs restored

Cyber-fraud losses, 2022–2026 (Rs Crore). Bars drawn to scale.



Task & investment scams surge

The fastest-growing cybercrimes in Karnataka (cases per year).



A Rs 15,000 loss becomes a Rs 1,50,000 disaster when victims pay the “withdrawal tax”.

WHY RECOVERY STAYS LOW DESPITE 87.4% FREEZE CAPABILITY

The 12-digit Bank UTR bottleneck 01

Operators cannot freeze on an app-specific ID (e.g. PhonePe’s `T261018…`). They need the **12-digit Bank UTR**. In panic, citizens lose **10–15 minutes** searching statements or quote the wrong ID.

Compound sunk-cost extortion 02

Scammers demand a secondary “withdrawal tax” (e.g. **30% of fake balances**). Desperate to recover money, victims pay again — turning small losses into ruin.

Digital Arrest coercion 03

Fake CBI / Mumbai Police on video call threaten arrest, demand isolation and “surety bail funds” to an “RBI verification account”. Citizens have **no instant way to verify** it is not real.

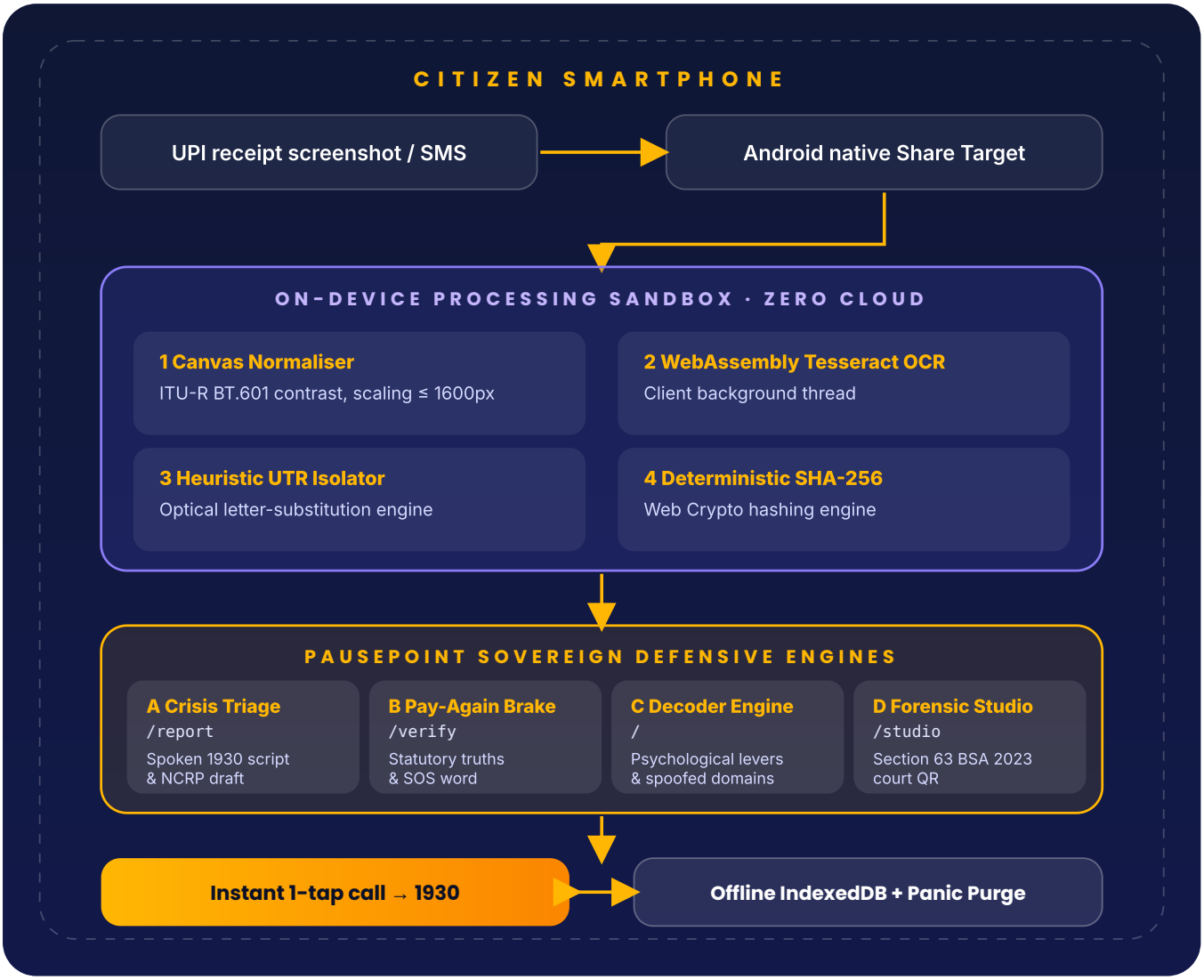
Cloud privacy distrust 04

Citizens fear uploading bank receipts and identity documents to unknown third-party cloud apps — and risking **further identity theft**.

03 · SOLUTION ARCHITECTURE

Client-Side Digital Sovereignty

Enterprise-grade cyber intelligence delivered directly on the user's phone, with **zero network transmission of financial data**. Everything below runs inside the browser sandbox.



Offline-first by design

A registered Service Worker caches every core route and WebAssembly asset — full crisis reporting works in airplane mode.

Nothing leaves the phone

OCR, threat parsing, hashing and storage all run locally. No receipt, ID or balance is ever transmitted.

Panic Purge

One tap wipes cached receipts, drafts and IndexedDB records if the victim feels watched or threatened.

0 bytes of financial data are transmitted off the device. That answers the **cloud privacy distrust** that stops frightened citizens from using other tools — and removes the data-protection liability altogether.

03.1 · FIVE CORE DEFENSIVE ENGINES

From panic to a report-ready docket

1 Emergency Crisis Triage & Instant 1930 Docket

`/report`

- **Zero-friction ingestion**
  Screenshots via file upload, camera, or the Android W3C Web Share Target API straight from PhonePe or Google Pay.
- **Optical substitution heuristics**
  Corrects UTR confusion ('I' read as '1', 'O' as '0') and stops account numbers, merchant names and balances being mistaken for the debit amount.
- **NCRP Ack parser**
  Reads the 14-digit acknowledgement SMS (Jan 2026 SOP format, starting with '3') and copies a pre-formatted complaint for `cybercrime.gov.in`.
- **Canvas preprocessing**
  Downscales to a 1600px ceiling to protect low-end RAM; applies ITU-R BT.601 luminance weighting and 1.25× contrast stretching.
- **Multi-payment fraud ledger**
  Aggregates multi-step task-scam transfers with running totals and per-transaction timestamps.
- **Instant 1-tap call**
  Dials 1930 directly from the generated docket.

AUTO-GENERATED SPOKEN 1930 SCRIPT

"Officer, I sent Rs 15,000 on 18 Oct at 3:30 PM via UPI to scammer handle xyz@upi. My debit Bank UTR is 429812345678 from my SBI account ending in 7788. Please freeze the destination account immediately."

2 "Before You Pay Again" Repeat-Payment Brake

`/verify`

Active cognitive interruption aimed at victims being extorted for secondary "unlock taxes" — three permanent statutory truths, shown at the moment of greatest pressure:

01
Legitimate employers NEVER ask employees to pay personal money to receive earned salary or task commissions.

02
Police, CBI and ED NEVER conduct "Digital Arrests", video trials or online bail verification over Skype or WhatsApp.

03
A UPI PIN or QR scan ONLY deducts money. It is technologically impossible to receive money by entering a PIN.

- **Trusted Family Shield & Safety Word**
  A pre-set family passphrase and 1-tap emergency WhatsApp alert defeat AI voice cloning and fake-kidnapping extortion.
- **Emergency 1-tap Panic Purge**
  Wipes all cached receipts, drafts and IndexedDB records in one tap (`⌘ ⌫`) if the victim feels watched or threatened.

Decode it. Prove it. Include everyone.

3 Psychological Manipulation Decoder

`/`

- **10-second decompilation**
  Scans pasted WhatsApp messages, SMS or email for manipulation triggers:

  Artificial urgency · Fear / threats · Fake authority · Secrecy / isolation

- **Counterfeit domain check**
  Cross-references links against authentic government portals (e.g. `echallan.parivahan.gov.in`, `incometax.gov.in`) and exposes spoofed look-alikes.

- **Interactive Cyber Attack Arena**
  Live simulations of four major Indian scam archetypes, letting users feel the cognitive pause first-hand:

  Task scams · Digital arrests · APK trojans · UPI PIN lures

4 Section 63 BSA 2023 Forensic Evidence Studio

`/studio`

Statutory court admissibility. Under Section 63 of the Bharatiya Sakshya Adhiniyam, 2023, electronic evidence needs proof of integrity from the point of creation.

- **Deterministic hashing**
  In-browser Web Crypto SHA-256 of the original receipt screenshot, timestamp and parsed fields.

- **Zero-dependency court QR stamp**
  An offline ISO/IEC 18004 QR matrix with GF(256) Reed-Solomon error correction, rendered directly to SVG — verifiable by officers or judges without proprietary software.

5 Sovereign Privacy & Inclusive Design

`PWA`

- **100% offline PWA shell**
  A Service Worker caches core routes and WebAssembly assets: full reporting in airplane mode or zero-connectivity zones.

- **Senior Citizen Mode**
  22px high-definition type, 64px touch-target floors and a calm, high-contrast palette.

- **Bilingual localisation**
  Complete Hindi (`hi-IN`) and English UI with calm, rate-controlled Web Speech voice playback.

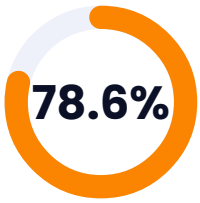
24/24 WCAG 2.2 AA audits passed · Zero critical / serious violations · Desktop + mobile viewports

04 · EMPIRICAL EVIDENCE & VALIDATION

Measured, not merely claimed

Campus field survey — N = 154

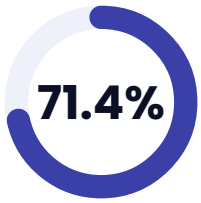
Students, faculty and senior citizens at JAIN (Deemed-to-be University).



received task-scam messages in the last 90 days



felt severe cognitive paralysis during cyber threats



did not know what a bank UTR was



of past victims faced secondary extortion demands

Paired timed reporting drill — 10 control vs 10 PausePoint

Mean time for 20 undergraduates to produce a usable 1930 script.



14 / 20 control participants reported wrong internal IDs

PausePoint: 100% UTR precision

EVIDENCE	SAMPLE / CORPUS	KEY FINDINGS
Real UPI testing corpus	40 real & synthetic receipts — Google Pay, PhonePe, Paytm, BHIM	**100%** 12-digit UTR recovery across all apps; corrected optical noise (`%` for `Rs` , `I` for `1` , `O` for `0`).
Unit test suite	99 tests across 18 suites (`npm test`)	**100%** passing in ~850 ms — canvas math, OCR heuristics, draft persistence, QR math, bilingual dictionary parity.
Playwright E2E suite	68 end-to-end tests	**100%** passing on desktop & mobile Chromium; **24/24** WCAG 2.2 AA audits, zero critical / serious violations.
Team pedigree	Human Firewall 2026 Hackathon	Core team won **3rd Prize** for the prototype psychological manipulation engine.

05 · ECONOMIC FEASIBILITY

Zero marginal cost. Infinite reach.

Because OCR, threat parsing and storage all run on the citizen's own phone, PausePoint needs **no cloud compute** — so it can protect millions across Karnataka and India without ongoing government server appropriations.

OPERATIONAL CATEGORY	INDUSTRY-STANDARD CLOUD APP	PAUSEPOINT GOLDEN HOUR
OCR compute	Cloud Vision API: ~Rs 120 per 1,000 scans	**Rs 0.00** Local WebAssembly on phone CPU
Threat parsing	Cloud server / LLM API: ~Rs 80 per 1,000 calls	**Rs 0.00** Local deterministic regex
User data storage	Cloud database: ~Rs 2,500 / month	**Rs 0.00** Sandboxed local IndexedDB
Edge CDN hosting	Cloudflare Pages / Vercel free tier	**< Rs 1,000 / month** $0–$10 even for millions of hits
Marginal cost per 100K users	~Rs 25,000 – Rs 50,000 / month	**Rs 0.00** Zero marginal-cost scalability

06 · IMPLEMENTATION & ROLLOUT ROADMAP

Campus → Police → National Digital Public Good

1

MONTHS 1–2

University pilot

Deploy emergency QR standees across JAIN campuses (FET, CMS, SGS), build it into first-year orientation, and spread it via student-union WhatsApp safety networks.

2

MONTHS 3–5

Police station & bank integration

Partner with Bengaluru City Police Cybercrime Division and Karnataka CID: QR posters in reception so complainants build a 1930 docket while queuing. Embed links in bank SMS-alert footers.

3

MONTHS 6+

National Digital Public Good

Open-source under a permissive licence for integration with `cybercrime.gov.in` and I4C Cyberdost channels. Expand to Kannada, Tamil, Telugu and Marathi.

Rs 0 marginal cost · **100% on-device** · deployable today. The only thing PausePoint needs from the state is **distribution** — not infrastructure.

07 · COMPLIANCE & REGULATORY ALIGNMENT

Built to fit the law, by design

NCRP Jan 2026 Standard Operating Procedure

Aligns intake fields precisely with the 14-digit acknowledgement number specification and the 1930 inter-bank lien protocols.

Bharatiya Sakshya Adhiniyam, 2023 — Section 63

Enforces cryptographic chain-of-custody for electronic records via deterministic SHA-256 evidence digests.

RBI Circular on Customer Protection

Helps citizens report unauthorised electronic transactions inside the critical 3-working-day window for zero-liability protection.

Digital Personal Data Protection Act, 2023

Eliminates privacy liability by design: 100% on-device sandboxing means no personal or financial data is collected, stored or processed on external servers.

08 · CONCLUSION

Replacing confusion with a 48-second brake

PausePoint Golden Hour bridges the critical operational gap between **citizen panic** and **law-enforcement inter-bank fund recovery**. By replacing confusion with an on-device, 48-second cognitive brake and an automated 1930 triage docket, it empowers ordinary Indian citizens to defend their life savings during the decisive Golden Hour.

Before PausePoint

The panicking victim...

- wastes 10–15 minutes hunting for the right ID
- quotes an app reference the helpline cannot use
- pays a second “withdrawal tax”
- watches the Golden Hour expire

With PausePoint

The same victim...

- has a report-ready script in 48.2 seconds
- reads the correct 12-digit Bank UTR aloud
- sees three statutory truths before paying again
- keeps every byte of data on their own phone

87.4%
freeze success inside the Golden Hour

48.2s
to a report-ready 1930 script

100%
UTR precision, 40-receipt corpus

0 bytes
of financial data leave the phone

TRY IT NOW · IT IS LIVE

Every second counts. Scan. Pause. Report.

PausePoint is not a slide-deck promise — it is a **working, production-hardened PWA** you can open on your phone right now, and a codebase you can read line by line.

Live web app

pausepoint.hkr.dpdns.org

Deployed on Cloudflare edge. Works offline once loaded.

Source code

github.com/himanshukr2245/23-sfic-pausepoint

99 unit tests · 68 Playwright E2E tests.

TEAM DELTA

HK — **Himanshu Kumar** — Lead Architect

NB — **Naga Bhavitha** — Team member

DC — **Deekshitha Chithaluru** — Team member

KV — **Komal Verma** — Team member

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Seva First Innovation Challenge (SFIC) 2026 · Theme 05: Surakshit Bharat (Digital Public Infrastructure & Cyber Security) · Track 1: Product / Technology Solution · Level 3 Open (Colleges & Universities)